

[11] Motor Control - Pinky

The schematic diagram illustrates the Motor Control Board's internal circuitry. The central component is the TB9051FTG,EL motor driver IC (U12). Its pins are connected as follows:

- VCC_P**: Connected to pin 2.
- +VBAT**: Connected to pin 11.
- GND**: Connected to pins 1, 3, 4, 6, 7, 10, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, and 29.
- EN_P**: Connected to pin 5.
- _OCM_P**: Connected to pin 8.
- PWMA_P**: Connected to pin 9.
- PWMB_P**: Connected to pin 10.
- OUT1_P**: Connected to pin 14.
- DIAG_P**: Connected to pin 18.
- OUT2_P**: Connected to pin 23.

Passive components include:

- R13** (1R 0.1% 0402) and **C56** (4.7uF 16V 0402) forming a network between +5VD and VCC_P.
- C57** (100nF 16V 0603) connected from VCC_P to GND.
- C58** (100nF 16V 0402) connected from +VBATD to GND.
- R14** (10K 0.1% 0402) and **C59** (470nF 16V 0402) forming a network between _OCM_P and OCM_P.
- R15** (510R 0.1% 0402) connected from OCM_P to GND.

A connector **J4** (BM09B-SRSS-TB) provides external access to various signals:

- GND**: Pin 1
- ECC_P**: Pin 2
- +3.3V**: Pin 3
- OUT1_P**: Pin 4
- GND**: Pin 5
- ECA_P**: Pin 6
- ECB_P**: Pin 7
- +3.3V**: Pin 8
- OUT2_P**: Pin 9

Comments:	Company: Willas 	Variant: DRAFT	Git Hash: e99a2fe	
	Board Name: Motor Control Board	Project Name: Master Hand		
Sheet Title:	File Name: Motor Control - Pinky.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
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